a) Advantages

i. It can be used for both AC and DC measurements.
ii. Robust and simple in construction.
iii. The cost is low.
iv. Possess high operating torque.
v. Less friction errors.
vi. These can withstand over loads momentarily.
vii. The range of instruments can be extended.

b) Disadvantages

i. This Scale is not uniform and is cramped at the lower end.
ii. There are serious errors due to hysteresis, frequency changes and stray magnetic fields.
iii. Power consumption is higher for low voltage range.
iv. There is a difference between AC and DC calibrations on account of the effect of inductance of the meter and eddy currents on AC.

6.5 Dynamometer type wattmeter

Dynamometer type watt meters are generally used for the measurement of power. It has two coils, one is a fixed coil and other is a moving coil. Both the fixed and moving coils are air cored. The fixed coil is divided into two equal portions in order to provide uniform field. The fixed coil is used as current coil and is connected in series with the load. The moving coil is used as pressure coil and is connected in parallel with the load. A high non-inductive resistance is connected in series with the moving coil to limit the current through it.

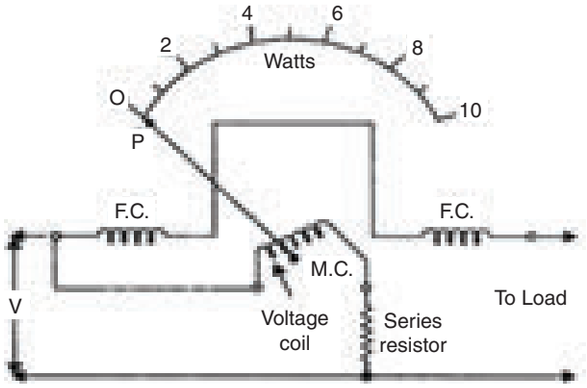


Fig. 6.9 Connection diagram of dynamometer type wattmeter

In this meter, the current coil carries the load current and the pressure coil carries the current proportional to the voltage across the circuit. The interaction of two magnetic fields produced by the current flowing through the fixed coils and moving coil causes the moving coil to turn about its axis. It is directly proportional to the product of voltage and current. The controlling torque is provided by the hair springs. These hair springs also lead the current into and out of the moving element. Air friction damping is used. Dynamometer type watt meters are suitable for both DC and AC power measurements.

a) Advantages

1. The scale is uniform.
2. High degree of accuracy can be obtained by careful design.
3. It is used both in AC and DC.

b) Disadvantages

1. The error due to the inductance of the pressure coil at low power factor is very serious.
2. Stray field may affect the reading of the instrument.

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